Experiment 23: Bayesian Network Construction and Visualization

Question

Problem Statement: A clinic wants a decision-support tool that reasons about the causes of a patient's symptoms under uncertainty. A medical diagnosis network with nodes Smoking, Pollution, Lung Disease, Cough, X-ray Result and Fatigue, with given conditional probability tables. The objective is to build a Bayesian network, define its CPDs and visualise its structure. Use pgmpy, NetworkX, Matplotlib.

Tasks:

1. Define the directed acyclic graph in pgmpy.
2. Specify the conditional probability distribution for every node.
3. Validate the model with check_model().
4. Draw the network with NetworkX.
5. List the conditional independencies and active trails in the network.

Code

```
"""Experiment 23: Bayesian Network Construction and Visualization (medical diagnosis)"""
import networkx as nx
import matplotlib.pyplot as plt
from pgmpy.factors.discrete import TabularCPD
try:
    from pgmpy.models import DiscreteBayesianNetwork as BayesianNetwork      # pgmpy >= 1.0
except ImportError:
    from pgmpy.models import BayesianNetwork                                  # older pgmpy

# ---------------- Task 1: define the DAG ----------------
model = BayesianNetwork([
    ("Smoking", "LungDisease"),
    ("Pollution", "LungDisease"),
    ("LungDisease", "Cough"),
    ("LungDisease", "XRay"),
    ("LungDisease", "Fatigue"),
])
print("Nodes:", list(model.nodes()))
print("Edges:", list(model.edges()))

# ---------------- Task 2: conditional probability distributions ----------------
# State order: row 0 = first state listed in state_names, row 1 = second state.
cpd_smoking = TabularCPD("Smoking", 2, [[0.7], [0.3]], state_names={"Smoking": ["no", "yes"]})
cpd_pollution = TabularCPD("Pollution", 2, [[0.8], [0.2]], state_names={"Pollution": ["low", "high"]})

# P(LungDisease | Smoking, Pollution): columns = (S,P) = (no,low), (no,high), (yes,low), (yes,high)
cpd_lung = TabularCPD(
    "LungDisease", 2,
    [[0.95, 0.85, 0.70, 0.50],          # LungDisease = no
     [0.05, 0.15, 0.30, 0.50]],         # LungDisease = yes
    evidence=["Smoking", "Pollution"], evidence_card=[2, 2],
    state_names={"LungDisease": ["no", "yes"], "Smoking": ["no", "yes"], "Pollution": ["low", "high"]})

cpd_cough = TabularCPD("Cough", 2, [[0.85, 0.20], [0.15, 0.80]], evidence=["LungDisease"], evidence_card=[2],
                       state_names={"Cough": ["no", "yes"], "LungDisease": ["no", "yes"]})
cpd_xray = TabularCPD("XRay", 2, [[0.95, 0.10], [0.05, 0.90]], evidence=["LungDisease"], evidence_card=[2],
                      state_names={"XRay": ["negative", "positive"], "LungDisease": ["no", "yes"]})
cpd_fatigue = TabularCPD("Fatigue", 2, [[0.80, 0.30], [0.20, 0.70]], evidence=["LungDisease"],
evidence_card=[2],
                         state_names={"Fatigue": ["no", "yes"], "LungDisease": ["no", "yes"]})

model.add_cpds(cpd_smoking, cpd_pollution, cpd_lung, cpd_cough, cpd_xray, cpd_fatigue)
for cpd in model.get_cpds():
    print(f"\nCPD of {cpd.variable}:\n{cpd}")

# ---------------- Task 3: validate the model ----------------
print("\nmodel.check_model() ->", model.check_model())

# ---------------- Task 4: draw the network with NetworkX ----------------
```

```
G = nx.DiGraph(model.edges())
pos = {"Smoking": (0, 2), "Pollution": (2, 2), "LungDisease": (1, 1),
       "Cough": (0, 0), "XRay": (1, 0), "Fatigue": (2, 0)}
colors = {"Smoking": "#fdae6b", "Pollution": "#fdae6b", "LungDisease": "#fc9272",
          "Cough": "#9ecae1", "XRay": "#9ecae1", "Fatigue": "#9ecae1"}
plt.figure(figsize=(8, 5.5))
nx.draw_networkx(G, pos, node_color=[colors[n] for n in G.nodes], node_size=3200, font_size=9,
                 edgecolors="k", arrowsize=25, width=1.8)
plt.title("Medical diagnosis Bayesian network (causes -> disease -> symptoms/test)")
plt.axis("off"); plt.tight_layout(); plt.show()


# ---------------- Task 5: independencies and active trails ----------------
print("\nAll conditional independencies in the network:")
print(model.get_independencies())


print("\nLocal independencies (node independent of its non-descendants given its parents):")
for node in model.nodes():
    li = model.local_independencies(node)
    if str(li).strip():
        print("  ", li)


print("\nActive trails (which nodes are connected / can influence each other):")
print("  No evidence, from Smoking            :", sorted(model.active_trail_nodes("Smoking")["Smoking"]))
print("  Observe LungDisease, from Smoking    :", sorted(model.active_trail_nodes("Smoking",
observed=["LungDisease"])["Smoking"]))
print("  Observe Cough, from Smoking          :", sorted(model.active_trail_nodes("Smoking",
observed=["Cough"])["Smoking"]))
print("  Observe LungDisease, from Cough      :", sorted(model.active_trail_nodes("Cough",
observed=["LungDisease"])["Cough"]))
print("\nInterpretation:")
print(" - Smoking and Pollution are independent a priori, but become dependent once LungDisease (or its
effect Cough/XRay) is observed (collider).")
print(" - Cough, XRay and Fatigue are conditionally independent of each other given LungDisease (common
cause).")
```

Output

```
Nodes: ['Smoking', 'LungDisease', 'Pollution', 'Cough', 'XRay', 'Fatigue']
Edges: [('Smoking', 'LungDisease'), ('LungDisease', 'Cough'), ('LungDisease', 'XRay'), ('LungDisease', 'Fatigue'), ('Pollution', 'LungDisease')]

CPD of Smoking:
+--------------+-----+
| Smoking(no)  | 0.7 |
+--------------+-----+
| Smoking(yes) | 0.3 |
+--------------+-----+

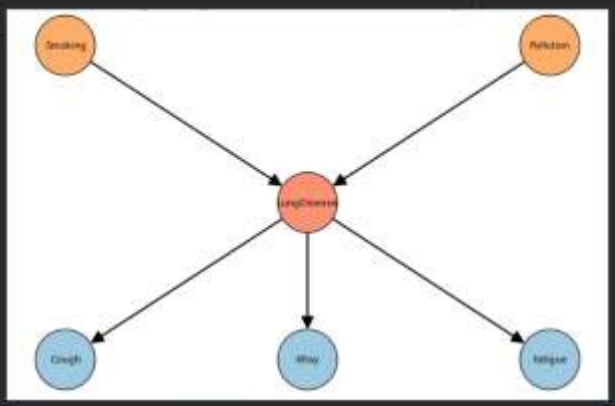
CPD of Pollution:
+-----------------+-----+
| Pollution(low)  | 0.8 |
+-----------------+-----+
| Pollution(high) | 0.2 |
+-----------------+-----+

CPD of LungDisease:
+-----------------+---------------+-----+---------------+---------------+
| Smoking         | Smoking(no)   | ... | Smoking(yes)  | Smoking(yes)  |
+-----------------+---------------+-----+---------------+---------------+
| Pollution       | Pollution(low)| ... | Pollution(low)| Pollution(high)|
+-----------------+---------------+-----+---------------+---------------+
| LungDisease(no) | 0.95          | ... | 0.7           | 0.5           |
+-----------------+---------------+-----+---------------+---------------+
| LungDisease(yes)| 0.05          | ... | 0.3           | 0.5           |
+-----------------+---------------+-----+---------------+---------------+

CPD of Cough:
+-------------+-----------------+-----------------+
| LungDisease | LungDisease(no) | LungDisease(yes)|
+-------------+-----------------+-----------------+
| Cough(no)   | 0.85            | 0.2             |
+-------------+-----------------+-----------------+
| Cough(yes)  | 0.15            | 0.8             |
+-------------+-----------------+-----------------+

CPD of XRay:
+---------------+-----------------+-----------------+
| LungDisease   | LungDisease(no) | LungDisease(yes)|
+---------------+-----------------+-----------------+
| XRay(negative)| 0.95            | 0.1             |
+---------------+-----------------+-----------------+
| XRay(positive)| 0.05            | 0.9             |
+---------------+-----------------+-----------------+

CPD of Fatigue:
+--------------+-----------------+-----------------+
```




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All conditional independencies in the network:
(Pollution ⊥ XRay | LungDisease)
(Fatigue ⊥ XRay | LungDisease)
(Pollution ⊥ Smoking)
(Cough ⊥ Fatigue | LungDisease)
(Cough ⊥ Pollution | LungDisease)
(Fatigue ⊥ Pollution | LungDisease)
(Cough ⊥ XRay | LungDisease)
(Smoking ⊥ XRay | LungDisease)
(Fatigue ⊥ Smoking | LungDisease)
(Cough ⊥ Smoking | LungDisease)

Local independencies (node independent of its non-descendants given its parents):
   (Smoking ⊥ Pollution)
   (Pollution ⊥ Smoking)
   (Cough ⊥ Fatigue, Pollution, Smoking, XRay | LungDisease)
   (XRay ⊥ Cough, Fatigue, Pollution, Smoking | LungDisease)
   (Fatigue ⊥ Cough, Pollution, Smoking, XRay | LungDisease)

Active trails (which nodes are connected / can influence each other):
  No evidence, from Smoking          : {'Cough', 'Fatigue', 'LungDisease', 'Smoking', 'XRay'}
  Observe LungDisease, from Smoking  : {'Pollution', 'Smoking'}
  Observe Cough, from Smoking        : {'Fatigue', 'LungDisease', 'Pollution', 'Smoking', 'XRay'}
  Observe LungDisease, from Cough    : {'Cough'}

Interpretation:
 - Smoking and Pollution are independent a priori, but become dependent once LungDisease (or its effect Cough/XRay) is observed (collider).
```